

Object Oriented Programming

- Complete Question Bank

(Chapter-wise)

Unit 1: Introduction to Object Oriented Programming

Long Answer Questions (10 marks)

1. What is object oriented programming? Explain objects, class, encapsulation, data hiding, inheritance, and polymorphism.
2. Why do we need object oriented programming? How can we use inheritance to reuse already written and tested code in programs? Discuss with suitable example.
3. What are the main features of the Object-Oriented Programming? Explain with suitable practical examples.
4. Discuss the features of Object-Oriented Programming. Differentiate between Object Oriented Programming and Procedural Oriented Programming.
5. Explain in detail the following principles of Object-Oriented Programming:
 - o
 - i. Data encapsulation and data hiding
 - o
 - ii. Inheritance and Polymorphism
 - o
 - iii. Abstraction

Short Answer Questions (5 marks)

1. Write any four features of object-oriented programming. Differentiate between operator overloading and function overloading.
 2. Discuss polymorphism with example.
 3. Explain the role of polymorphism in Object Oriented Programming.
 4. What is abstraction?
 5. What are the advantages and applications of OOP?
 6. How does OOP model real-world problems better than procedural programming?
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Unit 2: Basics of C++ Programming

Long Answer Questions (10 marks)

1. Why do we need the preprocessor directive `#include` ? Describe the major parts of a C++ program.
2. What do you mean by overloading of a function? When do we use this concept? Explain with example.

Short Answer Questions (5 marks)

1. What is namespace in C++? Explain with example.
2. Explain the purpose of a namespace with suitable example.
3. How can we define our functions inside the namespace and use them outside?
4. What is library function? How is it different from user defined function?
5. What is looping? Differentiate while loop with do-while loop with example.
6. Discuss the use of inline function with example.
7. What is inline function? Why is it used? Explain with example.
8. What is the principle reason for passing arguments by reference? Explain with suitable code.
9. What do you mean by passing arguments by reference? What is the importance of passing arguments by reference? Explain with example code.
10. What is function overloading? Explain with suitable example.
11. Explain the reason for using default arguments in the function by using suitable example.
12. What do you mean by dynamic initialization of variables?
13. Explain dynamic memory allocation in C++ using `new` and `delete` keyword.
14. Write the syntax and use of `getline()` and `write()` functions.
15. Explain with examples various manipulators used in C++.
16. What is type casting? Explain with suitable example.
17. Why type conversion is necessary in OOP? Explain with example, the type conversion routine.
18. Explain about keywords, identifiers, data types in C++, `enum`, operators, scope resolution operator.

Programming Questions (5 marks)

1. Write a program that shows an example of function overloading.
2. Write a program to find the square of given integer using inline function.
3. Write a program to find the cube of given integer using inline function.
4. Write a program to find the square root of given integer using inline function.
5. Write a program to convert centigrade into Fahrenheit temperature.
6. Write a program to convert feet into meter.
7. Write a program to convert inch into centimeter.
8. Write a program in C++ to count a number of words in a line of text.

Unit 3: Classes & Objects

Long Answer Questions (10 marks)

1. Discuss features of class and object. Design a class to represent a bank account with data members name, account-number, account-type, and balance and functions to assign initial values, to deposit an amount, to withdraw an amount after checking balance, and to display the name and balance.
2. Why constructor and destructor are required in Object Oriented Programming? Explain with suitable example.
3. Explain the role of constructor and destructor in Object Oriented Programming. Discuss user defined parameterized constructor with suitable example.
4. What is constructor? Explain their types? Discuss user defined parameterized constructor with suitable example.
5. Define a student class (with necessary constructors and member functions) in Object Oriented Programming. Derive a Computer Science and Mathematics class from student class adding necessary attributes. Use these classes in a main function and display the average marks of computer science and mathematics students.

Short Answer Questions (5 marks)

1. What is class and object? Explain class and object with suitable examples.
 2. What is constructor? Differentiate it with destructor.
 3. Why constructor is needed? Explain different types of constructors with example.
 4. What are the characteristics of constructor?
 5. What is destructor?
 6. Explain about accessing a member function outside a class with example.
 7. How can you define member function outside of a class? Explain with example.
 8. Why dynamic object is needed? Explain with suitable example.
 9. How can you classify objects? Why dynamic objects are needed?
 10. What is static variable and static methods? Why is it important to use static data members?
 11. Example showing the real use of static data members with brief explanation.
 12. What is the use of "this" pointer? Explain with example.
 13. Explain about "this" pointer with suitable example.
 14. Explain encapsulation in details with example program.
 15. What are the various access specifiers used in C++? Explain.
 16. Explain the different types of class access specifiers.
 17. Differentiate between structure and class in terms of access modifier.
 18. Differentiate between private, public and protected variables with suitable example.
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Unit 4: Operator Overloading

Long Answer Questions (10 marks)

1. What is operator overloading? What are the benefits of overloading? How is operator overloading different from function overloading? Write a program that shows an example of function overloading.
2. Explain operator overloading. Write a program that overloads insertion and extraction operators.

Short Answer Questions (5 marks)

1. What is operator overloading? Explain their types with suitable examples.
 2. Write a program that illustrates the conversions between objects of different classes having conversion function in source object.
 3. Write a program that decreases an integer value by 1 (one) by overloading – operator.
 4. Write a program to compute subtraction of two complex numbers using operator overloading.
 5. Unary operator overloading using member function (concept with program)
 6. Unary operator overloading using friend function (concept with program)
 7. Binary operator overloading using member function (concept with program)
 8. Binary operator overloading using friend function (concept with program)
 9. Extraction operator overloading (concept with program)
 10. Insertion operator overloading (concept with program)
 11. Basic to user-defined data conversion (concept with program)
 12. User-defined to basic data conversion (concept with program)
 13. User-defined to user-defined data conversion (implicit conversion/conversion on destination class)
 14. User-defined to user-defined data conversion (explicit conversion/conversion on source class)
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Unit 5: Inheritance

Long Answer Questions (10 marks)

1. What is inheritance? Explain the ambiguities associated with multiple inheritance with suitable example programs.
2. What is inheritance? Why is it important?
3. What is Inheritance? Explain their types with their suitable examples.

Short Answer Questions (5 marks)

1. What are the types of inheritance? Explain with examples.
 2. Explain the difference between private and public inheritance with suitable diagram.
 3. Differentiate Public and Private Inheritance.
 4. Differentiate between base class and derived class with suitable examples.
 5. Differentiate Derived Class and Base Class.
 6. Differentiate between super class and sub class with suitable examples.
 7. Why ambiguity occurs in multiple inheritance? What are the ways to solve it? (Explain in detail with program)
 8. Ambiguity in multiple inheritance and way to solve it.
 9. Explain about constructor and destructor in derived class.
 10. Derived class constructor, Default and parameterized constructor.
 11. What is aggregation? Explain with example.
 12. Differentiate container class from inheritance. Explain with suitable example.
 13. What is container class? Differentiate container class from inheritance.
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Unit 6: Virtual Function, Polymorphism, and Miscellaneous C++ Features

Long Answer Questions (10 marks)

1. How late binding is different from early binding. Write a program that explains late binding using virtual function.

Short Answer Questions (5 marks)

1. What is virtual function? Explain.
 2. Explain the concept of virtual functions.
 3. Explain the role of virtual function in Object Oriented Programming.
 4. What is polymorphism? Explain compile-time and run-time polymorphism with examples.
 5. Differentiate between early binding and late binding / static binding and dynamic binding / compile-time polymorphism and run-time polymorphism.
 6. Write a program to illustrate late binding using virtual function.
 7. Differentiate between abstract base classes and concrete classes.
 8. What is abstract class? Explain with example.
 9. Explain the concept of virtual destructor.
 10. Differentiate between virtual and pure virtual function with suitable example.
 11. Why friend function is required? Discuss with example.
 12. What is friend function? Why it is used in OOP? Explain with an example.
 13. What is friend function? Why friend function is used? Explain with example. Why it is against the philosophy of encapsulation or data hiding?
 14. Explain the friend function with its syntax.
 15. Differentiate between overriding vs overloading.
 16. Differentiate between function overriding and function overloading. Explain with suitable example.
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Unit 7: Function Templates and Exception Handling

Long Answer Questions (10 marks)

1. Why do we need exceptions? Explain "exceptions with arguments" with suitable program.

Short Answer Questions (5 marks)

1. What is function template? Differentiate it with class template.
2. Explain the function templates with example.

3. Why exception handling is required? Explain with suitable example.
 4. Write a program to show exception handling.
 5. Write a C++ program containing a possible exception. Use a try block to throw it and a catch block to handle it properly.
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Unit 8: File Handling and Streams

Short Answer Questions (5 marks)

1. What are the advantages of using the stream classes for I/O? Write a program that writes object to a file.
 2. Write the syntax and use of `getline()` and `write()` functions.
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Additional Important Programming Questions

Class Design Programs

1. Define a Shape class (with necessary constructors and member functions). Derive Triangle and Rectangle classes from Shape class adding necessary attributes. Use these classes in a main function and display the area of triangle and rectangle.
2. Define a Clock class (with necessary constructors and member functions). Derive a wall_clock class from clock class adding necessary attributes. Create two objects of wall_clock class with all initial state to 0 or NULL.
3. Design a class to represent a bank account with appropriate data members and member functions.

Conversion Programs

1. Temperature conversion programs (Celsius to Fahrenheit, etc.)
2. Unit conversion programs (feet to meter, inch to centimeter)
3. Mathematical operation programs using inline functions

Operator Overloading Programs

1. Complex number arithmetic using operator overloading
 2. Unary operator overloading (++ , -- , etc.)
 3. Binary operator overloading (+ , - , * , / , etc.)
 4. Stream operator overloading (<< and >>)
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Exam Strategy Tips

For Long Answer Questions (10 marks):

- Always include definition, explanation, and example/program
- Include advantages and applications where relevant

For Short Answer Questions (5 marks):

- Include small code examples where asked

Common Topics to Focus On:

1. OOP principles and features
2. Constructor and destructor concepts
3. Inheritance types and access specifiers
4. Operator overloading (all types)
5. Virtual functions and polymorphism
6. Exception handling basics
7. Friend functions and their usage
8. Type conversion mechanisms